

Graphing quadratic functions



1 Consider the function $y = x^2$.

a Complete the following table of values.

x	-3	-2	-1	0	1	2	3
y							

b Plot the points in the table of values on a coordinate plane.

c Hence plot the curve.

d Are the y values ever negative?

e Write down the equation of the axis of symmetry.

f State the minimum y value.

g For every y value greater than 0, how many corresponding x values are there?

2 Consider the function $y = -x^2$.

a Complete the following table of values.

x	-3	-2	-1	0	1	2	3
y							

b Plot the points in the table of values on a coordinate plane.

c Hence plot the curve.

d Are the y values ever positive?

e State the maximum y value.

f Write down the equation of the axis of symmetry.

3 For the following quadratic functions:



i Complete the following table of values.

x	-3	-2	-1	0	1	2	3
y							

ii Graph the function.

- a $y = 3x^2$ b $y = -2x^2$ c $y = 3x^2 + 3$ d $y = 3x^2 - 3$
e $y = -2x^2 + 5$ f $y = -2x^2 - 2$

4 For the following quadratic functions:

i Complete the following table of values. ii Graph the function.

x	-2	-1	0	1	2
y					

- a $y = \frac{1}{2}x^2$ b $y = -\frac{1}{2}x^2$

5 Consider the function $y = (x - 2)^2$.

a Complete the following table of values.

x	0	1	2	3	4
y					

- b Sketch the graph of the function.
c State the minimum y value.
d State the x value that corresponds to this minimum y value.
e State the coordinates of the vertex.

6 Consider the function $y = -(x - 3)^2$.



a Complete the following table of values.

x	1	2	3	4	5
y					

b Sketch the graph of the function.

c State the maximum y value.

d State the x value that corresponds to this maximum y value.

e State the coordinates of the vertex.

7 For the following functions:

i Complete the following table of values.

x	-3	-2	-1	0	1
y					

ii Sketch the graph of the function.

iii State the minimum y value.

iv State the x value that corresponds to this minimum y value.

v State the coordinates of the vertex.

a $y = (x + 1)^2$

b $y = -(x + 1)^2$

8 The graph of $y = x^2 + 6$ has no x -intercepts. Is this statement true or false?

9 Consider the equation $y = x^2 + 4$.

a Complete the set of solutions for the given equation.

$A(-3, 1)$, $B(-2, 1)$, $C(-1, 1)$, $D(1, 4)$, $E(1, 1)$, $F(2, 1)$, $G(3, 1)$

b Plot the points C , D and E on the coordinate plane.

c Plot the curve that results from the entire set of solutions for the equation being graphed.



10 Consider the equation $y = x^2 + 6x + 5$.

- a** Complete the set of solutions for the given equation.
 $A(-5, 1), B(-2, 1), C(-6, 1), D(-1, 1), E(0, 1), F(-3, 1), G(-4, 1)$
- b** Plot the points on the coordinate plane.
- c** Plot the curve that results from the entire set of solutions for the equation being graphed.

11 Find the maximum value of y for the quadratic function $y = -x^2 + 10x - 25$.

12 Does the curve $y = -x^2 + 3x - 10$ has a minimum value or a maximum value?

13 Consider the curve $y = -2x^2 - 4x + 10$.

- a** Find the axis of symmetry.
- b** State the maximum value of y .
- c** State the coordinates of the vertex.
- d** Graph the function using the maximum point on the curve.

14 Consider the curve $y = x^2 + 6x + 4$.

- a** Find the axis of symmetry.
- b** State the minimum value of y .
- c** Graph the function using the minimum point on the curve.

15 For the following quadratic functions:

- i** Find the axis of symmetry.
- ii** Find the vertex of the parabola .
- iii** State the y value of the y -intercept of this quadratic function.
- iv** Sketch the graph of the function.

a $y = 3x^2 - 6x + 8$

b $y = -2x^2 + 4x - 7$



- 16 Vincent is training for a remote control plane robotics competition. He wants to fly the plane along the path of a parabola, and so has chosen the equation $y = 3x^2$, where y is the height in meters of the plane from the ground, and x is the horizontal distance in meters of the plane from its starting point.

a Complete the following table of values of the height and distance.

x (m)	0	1	2	3	4
y (m)					

- b Plot the shape of the path of the plane.
- c State the lowest height of the plane.
- d State the x value that corresponds to this minimum y value.
- e State the coordinates of the vertex.

- 17 Consider the quadratic functions, $f(x)$ and $g(x)$ that are shown:
Which function has the smallest minimum value? State its value.

- | | | | | | | |
|--------|----|----|----|----|----|----|
| x | -6 | -5 | -4 | -3 | -2 | -1 |
| $f(x)$ | -8 | -9 | -8 | -5 | 0 | 7 |
- $g(x) = x^2 - 10x + 9$